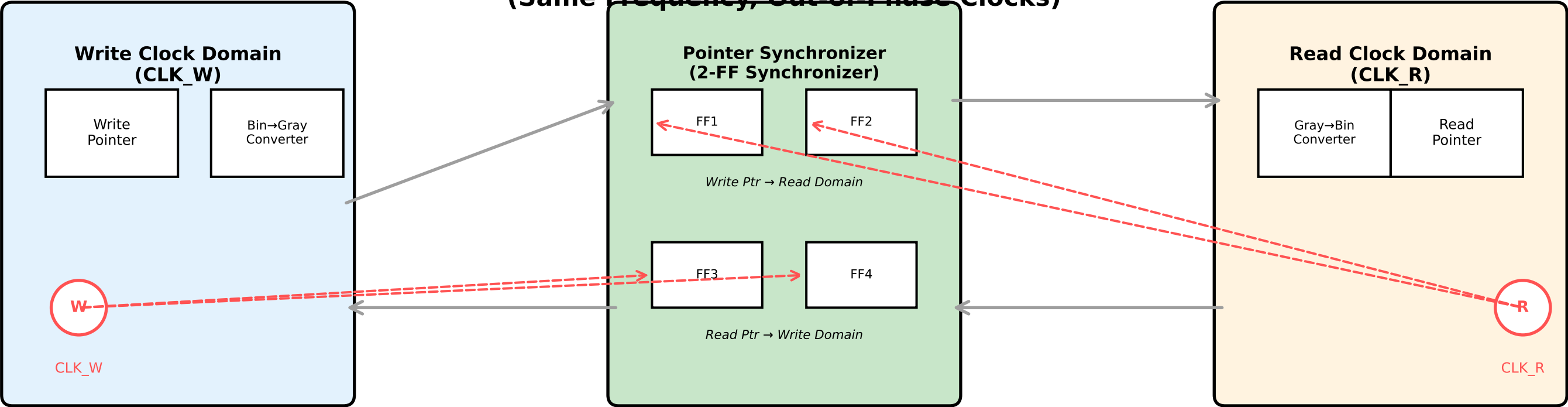


Asynchronous FIFO Pointer Synchronizer Schematic (Same Frequency, Out-of-Phase Clocks)



Write Pointer (Gray)

Read Pointer (Gray)

CLK_W (Same Freq, Out of Phase)



CLK_R (Same Freq, Out of Phase)



- Key Features:
1. Gray Code encoding ensures only 1-bit changes per transition
 2. Dual Flip-Flop Synchronizer (2-FF) eliminates metastability
 3. Same frequency, out-of-phase clocks: Same freq but different phase
 4. Write pointer synchronized to read domain, read pointer to write domain
 5. Synchronized pointers used for empty/full detection